

Perpendicular lines



- 1 Describe what it means for two lines to be perpendicular.
- 2 If two lines are perpendicular, state the product of their gradients.
- 3 State whether the following pairs of lines are perpendicular:
 - a $y = x + 1$ and $y = x - 1$
 - b $y = 3x - 7$ and $y = -3x + 6$
 - c $y = -3x + 6$ and $y = \frac{x}{3} + 7$
 - d $y = \frac{2x}{3} + 4$ and $y = -1 - \frac{3x}{2}$
 - e $y = -\frac{x}{3} - 9$ and $y = -(3x - 27)$
 - f $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$
 - g $y = -\frac{2x}{3} + 1$ and $y = -1 + \frac{3x}{2}$
 - h $y = \frac{3x}{5} + 6$ and $y = \frac{-5x - 30}{-3}$
- 4 Given that the following pairs of lines are perpendicular, calculate the value of m :
 - a $5x + 4y + 8 = 0$
 $mx + 9y - 8 = 0$
 - b $-6x + 6y - 8 = 0$
 $mx + 10y + 8 = 0$
- 5 Find the gradient of the line perpendicular to the following lines:
 - a Line with gradient 6
 - b $y = 6x + 3$
 - c $y = \frac{x}{8} + 3$
 - d $y = 8 - x$
- 6 Consider the lines $L_1: 8x = -7y + 3$ and $L_2: 7y - 8x - 7 = 0$.
 - a Find the gradient of line L_1 .
 - b Find the gradient of line L_2 .
 - c Are the two lines perpendicular?
- 7 Find the equation of a line described by the following information:
 - a Perpendicular to the x -axis and passes through $(-8, -1)$.
 - b Perpendicular to the y -axis and passes through $(-8, -8)$.
 - c Perpendicular to $y = -\frac{x}{2} + 5$, and goes through the point $(0, 6)$.
 - d Perpendicular to $y = 6x + 10$, and has the same y -intercept.
 - e Perpendicular to the line that passes through $(1, 1)$ and $(3, 13)$, and has a y -intercept of $(0, -4)$.
- 8 The line L_1 is perpendicular to $y = 5x - 4$ and cuts the y -axis at 1.

a Find the gradient of line L_1 .



b Find the equation of line L_1 .

9 The line L_1 is perpendicular to $y = 10x + 8$ and passes through the point of intersection of the lines $y = x + 5$ and $8x - 10y + 60 = 0$.

a Find the gradient of line L_1 .

b Find the point of intersection of $y = x + 5$ and $8x - 10y + 60 = 0$.

c Find the equation of the perpendicular line L_1 .

Applications

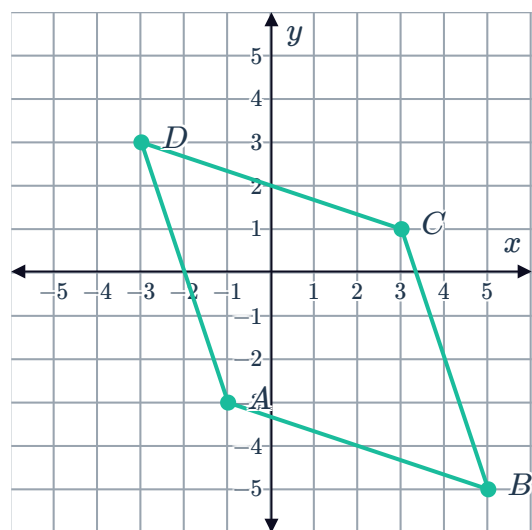
10 Consider the rhombus $ABCD$ on the number plane:

a Find the gradients of its diagonals.

i Gradient of AC

ii Gradient of BD

b Are the diagonals of the rhombus perpendicular? Explain your answer.



11 The four vertices of quadrilateral $ABCD$ have been plotted on the number plane:

a Find the gradient of the following sides:

i AB ii BC iii DC iv AD

b Given that side $AB = BC$, determine the type of quadrilateral described by the four points.

